

ARCHITECTURE COMPLIANCE AUDIT • UPDATE PASS

# Two tasks landed. The organism boots and remembers itself.

Re-comparison of spawn/ against Architecture.md, measured against the 7/2 6:31pm audit. Covers Task #16 (Composition Root) and Task #17 (Replay-on-Start) — everything else in the codebase is unchanged since the last pass.

BASELINE AUDIT 7/2/26 6:31 PM    THIS PASS 7/2/26 10:40 PM

READ-ONLY • NO CODE MODIFIED BY THIS AUDIT

|                    |               |                  |
|--------------------|---------------|------------------|
| OVERALL COMPLETION | TESTS PASSING | COMPOSITION ROOT |
|--------------------|---------------|------------------|

~~53%~~ → **59%**

~~134/134~~ → **177/177** not wired → **wired**

COMPONENTS FULLY  
COMPLETE

~~2/8~~ → **2/8**

## What shipped

Both tasks were scoped as pure Kernel/composition work — no component's internal logic (Perception, World Model, Executive, Governor, Executor, Memory & Ledger, Inference Port) changed. Their scores below are carried forward unchanged.

### Task #16 — Composition Root

roadmap item 6

- `main.py` now constructs exactly one `Kernel` and exactly one instance of all seven components against it, plus the Inference Port and its provider registry.
- Added a deterministic bootstrap path: one observation runs the full `Observation → Belief → Opportunity → Plan → Approval → Execution → Outcome → Memory` chain through typed events only — no component called directly out of sequence.
- 10 new integration tests assert single shared Kernel, exactly-once subscriptions, and the full 18-event cascade in architectural order.

## Task #17 — Replay-on-Start

roadmap item 3

- `Kernel.replay()` re-dispatches every persisted event through the same subscriber path live events use — zero component-specific replay code.
- `Kernel.start()` now runs replay automatically, exactly once, before the first `KERNEL_STARTING` event — replay is structurally guaranteed to finish before normal scheduling begins.
- Nested `publish()` calls are suppressed during replay, so re-running old handlers reconstructs state without appending a single duplicate event to the log.

- 7 new tests cover empty-log boot, full state reconstruction after a simulated crash, event-order preservation, determinism across repeated restarts, and zero duplication.

# Completion by component

Same weighting as the baseline audit: each named sub-responsibility scored complete / partial / missing, percentage is the component average.

| COMPONENT   | BEFORE | TREND | AFTER | Δ   |
|-------------|--------|-------|-------|-----|
| Kernel      | 17%    |       | 67%   | +50 |
| Perception  | 83%    |       | 83%   | —   |
| World Model | 100%   |       | 100%  | —   |

| COMPONENT       | BEFORE | TREND | AFTER | Δ |
|-----------------|--------|-------|-------|---|
| Executive       | 30%    |       | 30%   | — |
| Governor        | 63%    |       | 63%   | — |
| Executor        | 25%    |       | 25%   | — |
| Memory & Ledger | 6%     |       | 6%    | — |
| Inference Port  | 100%   |       | 100%  | — |

# Kernel, in detail

The only component whose score moved. Four of six named responsibilities are now complete — up from one.

| RESPONSIBILITY      | BEFORE   | AFTER     | NOTE                                                                                                                                               |
|---------------------|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| event log semantics | complete | unchanged | Already complete at baseline.                                                                                                                      |
| scheduler           | missing  | complete  | FIFO Scheduler now owns queueing<br>Kernel delegates<br>publish/process_next/run_until<br>to it. Landed just before this pass, at<br>of #16/#17.   |
| timers              | missing  | missing   | Scheduler is pure FIFO — no “fire at,<br>T” primitive yet. Belief decay is still in<br>manually. Roadmap item 5, second h                          |
| replay              | missing  | complete  | Task #17. Kernel.replay() dispatch<br>persisted events through live subscri<br>handlers, exactly once each, before<br>start() admits any new work. |
|                     |          |           |                                                                                                                                                    |

| RESPONSIBILITY | BEFORE  | AFTER    | NOTE                                                                                                                                                                                                                                                                                       |
|----------------|---------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| crash recovery | missing | complete | Kill the process, reopen the same EventLog path, rebuild the organism start() — verified deterministic across repeated restarts by test_replay.py. Governor's constitution/budget are supplied as boot config, not replayed were never event-sourced — see Governor's own gap, unchanged). |
| snapshotting   | missing | missing  | data/snapshots/ is still an empty s<br>Replay currently means “re-read the log from event zero” — fine at today volume, roadmap item 4.                                                                                                                                                    |

# Architecture-wide checks

| CHECK       | BEFORE  | AFTER    | NOTE                                                                                                                                                                                        |
|-------------|---------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Persistence | missing | complete | EventLog is JSONL on disk per append. Component (BeliefStore, PlanStore, ...) purely in memory by design no longer need their own persistence, because repo reconstructs them from the log. |
| Recovery    | missing | complete | Same root cause as persistence now closed by the same <code>main.build_organism()</code> accepts a pre-wired Kernel reboot can point at the same path.                                      |
|             |         |          |                                                                                                                                                                                             |



| CHECK            | BEFORE                                                  | AFTER     | NOTE                                                                                                       |
|------------------|---------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------|
| Composition root | missing                                                 | complete  | Task #16. First point the v<br>organism is provably alive<br>unit test.                                    |
| Determinism      | no<br>violations,<br><i>replay</i><br><i>unverified</i> | complete  | Replay-determinism now<br>tested across three sequ<br>restarts on the same log.<br>caveat below.           |
| Auditability     | partial                                                 | unchanged | correlation_id is still l<br>every event — roadmap<br>cheapest item on the bo<br>not picked up by either t |
| Missing events   | partial                                                 | unchanged | PlanAbandonedEvent /<br>ExecutiveDecisionEve<br>defined, still never publis                                |

| CHECK                          | BEFORE | AFTER | NOTE                                                                                                                                   |
|--------------------------------|--------|-------|----------------------------------------------------------------------------------------------------------------------------------------|
|                                |        |       | timer-fired event still exists (still missing).                                                                                        |
| Ownership & cross-store access | clean  | clean | Held through both tasks now imports every component that's the composition record components still only imported src.events and src.ke |

# A finding from building Task #17

Reconstructed state is identical to uninterrupted execution for everything the architecture actually asks to be identical — belief confidence, plan/approval counts, budget deltas, outcome success, ledger entries. It is *not* identical for internal record IDs: Executor, Executive, Governor, and Memory & Ledger each mint a fresh `uuid4()` inside the handler on every run rather than deriving it from the event, so

`action_id` and friends differ between an original run and its replay. Not a defect — those IDs were never part of the durable identity axiom — but worth knowing before anyone writes a test (or a downstream feature) that assumes replay reproduces IDs byte-for-byte.

## Roadmap: where the 18-item board stands

01 **Populate `correlation_id` on every publish** **not started**

Still the cheapest open item. Neither #16 nor #17 touched it.

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02 **Kernel: disk persistence for the event log** **done**

Landed ahead of this pass, immediately unblocked #3.

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03 **Kernel: replay-on-start** **done – Task #17**

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04 **Kernel: snapshotting** **not started**

Log is small; not urgent yet, but grows unbounded until this exists.

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## 05 Kernel: scheduler + timer table **half done**

FIFO scheduler shipped just before this pass. The timer half ("fire at/after T") is still open.

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## 06 Wire the composition root **done – Task #16**

**Items 07–18 unchanged** — still open, in original order: Perception normalization adapters, Executive/Governor capital & attention allocation against live budget, Executive → Inference Port wiring for EV estimation, Executive ranking & plan selection, Governor rule engine + owner-approved amendment path, Governor real escalation path, Executor sandboxing, Executor retries, Executor credentials, Memory & Ledger prediction tracking, medium learning loop, slow learning loop. None of these were in scope for #16 or #17, and none regressed.

spawn/ · read-only audit · no code modified      177/177 tests passing at time of this audit